Amazon Product Analysis Dashboard

Objective:
To analyze Amazon product data to understand pricing, discounts, and customer ratings across different categories.

Dataset:
The dataset contains Amazon product information including category, pricing, discounts, ratings, and user engagement data (rating count).
Source: https://www.kaggle.com/datasets/karkavelrajaj/amazon-sales-dataset

Data Cleaning:

- Removed unnecessary columns
- Handled missing values
- Converted price columns to numeric
- Extracted main category from category column for better analysis
- Created new features like popularity and rating category features

Analysis:

- Category-wise popularity using rating count
- Average price and discount analysis
- Rating distribution (High vs Low)
- Top 10 most popular products

Insights:

- Electronics & Accessories are most popular
- Higher discounts don’t guarantee higher ratings
- Few products dominate user engagement

Sum of popularity
Category vs Popularity
main_category
Toys&Games
OfficeProducts
MusicalInstruments
HomeImprovement
Home&Kitchen
Health&PersonalCare
Electronics
Computers&Accessories
Car&Motorbike
0
10000
Thousands
Total
Average of discounted_price
Average Price by Category
7000
6000
5000
4000
3000
2000
1000
0
Electronics
Car&Motorbike
Home&Kitchen
Health&PersonalCa...
Computers&Access...
MusicalInstruments
HomeImprovement
OfficeProducts
Toys&Games
Total
main_category
Count of rating_category
Rating Distribution
rating_category
High
low
rating_category
High
low
Sum of popularity
Top 10 poducts
product_name
Redmi 9A Sport (Carbon...
Redmi 9A Sport (Coral...
Redmi 9 Activ (Carbon...
Redmi 9A Sport (Coral...
boAt BassHeads 100 in-...
boAt Bassheads 100 in...
boAt Bassheads 100 in...
JBL C100SI Wired In Ear...
Amazon Basics High-...
Amazon Basics High-...
AmazonBasics Flexible...
0
400000
800000
Total
Average of discount_percentage
Discount by Category
main_category
Toys&Games
OfficeProducts
Home&Kitchen
Car&Motorbike
MusicalInstruments
Computers&Accessories
HomeImprovement
0%
20%
40%
60%
80%
Total
AutoSort
main_category: Largest to Smallest by Average of discount_percentage
Average of rating
Average Rating by Category
4.4
4.3
4.2
4.1
4
3.9
3.8
3.7
3.6
3.5
Electronics
Health&PersonalCa...
Home&Kitchen
HomeImprovement
MusicalInstruments
OfficeProducts
Toys&Games
Total
main_category
main_category
Car&Motorbike
Computers&Access...
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